STAFF REVIEW · H446 130 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A shorter representation for a repeated substring.

2. Linear/sequential search in the worst case.

3. The left subtree.

4. First in, first out.

I Do — source worked steps

1. Alice: message + key K — Predict the first step from the prompt.

2. Encrypted ciphertext — Explain the intermediate result or decision.

3. Bob: decrypt with K — A secure symmetric cipher transforms plaintext using key K. Bob decrypts with the same key; an interceptor without K should not recover plaintext feasibly. Security requires a strong cipher and protected key.

We Do — guided CHECK

1. Sensitive backup
Predict the first step from the prompt.
Encrypt with shared key
Explain the intermediate result or decision.
Protect key separately
Symmetric algorithms are efficient for encrypting large files, but the encryption key must be stored and shared securely; encrypting a file is not by itself proof of who created it.

You Do — source mark schemes

1. Readable input [2], secret key used by encrypt/decrypt [2], ciphertext unreadable without key [2].

2. Both need secret [2], secure initial transfer/storage [2], compromise exposes communications or data [2].

3. Bulk speed [2], confidentiality [2], safe key management/recovery [2], practical limitations and conclusion [2].

Exit Ticket — source answer

1. False. Those are features of asymmetric encryption.